

# Engineering Graphics Using CAD Software

A full laboratory handbook for learning CAD tools, precision drafting, six foundational figures, four isometric drawings, two complete 2D engineering drawings, plotting, lab examination workflow and open-ended project reporting.

Course Code

**2038**

Programme

**Electrical & Electronics / Renewable Energy / Electrical & EV Technology**

Semester

**2**

Credits

**No Credit**

Category

**Engineering Science**

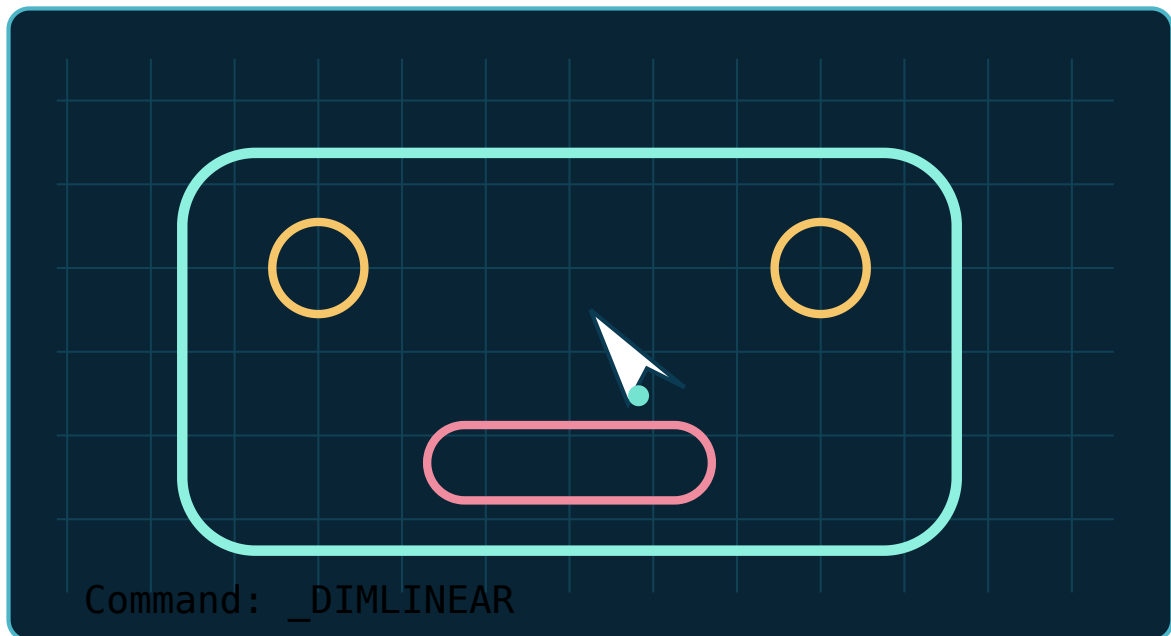
Periods

**3 practical/week • 45/semester**

## How this handbook matches the official syllabus

The official syllabus specifies basic CAD tools, at least six CAD figures, at least four isometric figures, at least two complete two-dimensional figures, two three-hour lab examinations and a 15-mark open-ended group activity. Every one of those requirements is built into this handbook. Verified again against the uploaded three-page official syllabus: programme, no-credit status, 45 practical periods, CO-PO mapping, lab-exam split, references, online resources and the open-ended experiment requirement are reflected here.





## Official practical path

Identify tools and apply them to sketches.

Develop at least six CAD figures.

Develop at least four isometric figures.

Construct at least two complete 2D figures.

LINE OFFSET TRIM ISO

# CAD converts engineering ideas into precise, editable drawings

Manual drawing develops visualisation; CAD adds speed, accuracy, revision control, layers, reusable geometry and professional plotting. Electrical and renewable-energy students use CAD for layouts, mounting plates, panel drawings, cable routes, component details and equipment arrangement.

## Accuracy

Coordinates, object snaps and constraints reduce measurement and alignment errors.

## Editability

MOVE, COPY, OFFSET, TRIM and grips modify geometry without redrawing the entire sheet.

## Communication

Layers, dimensions, line types, text and plotting turn geometry into an engineering document.

## How to use this handbook

### 1. Set up

Study units, workspace, coordinates, layers and object snaps before drawing.

### 2. Observe

Follow the command sequence and the geometric reason behind each step.

### 3. Draw

Complete every minimum syllabus figure and save each file with a clear name.

### 4. Verify

Check dimensions, symmetry, layer use, line type, title block and plot preview.

### Lab exam habit

Read the complete question first. Plan layers and construction lines, save immediately, then draw. Use object snaps instead of visual guessing. Dimension only after geometry is complete and verified.

# Open any handbook section

## OFFICIAL SYLLABUS ANALYSIS

## Course identity, objectives and prerequisites

|                              |                                                                                                                             |                             |                     |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------|
| <b>Programme</b>             | Diploma in Electrical and Electronics Engineering / Renewable Energy / Electrical Engineering & Electric Vehicle Technology | <b>Semester</b>             | 2                   |
| <b>Course Code</b>           | 2038                                                                                                                        | <b>Credits</b>              | No Credit           |
| <b>Official Course Title</b> | Engineering Graphic using CAD software                                                                                      | <b>Course Category</b>      | Engineering Science |
| <b>Periods per week</b>      | 3 (L:0, T:0, P:3)                                                                                                           | <b>Periods per semester</b> | 45                  |

### Course Objective 1

To use CAD software for drafting and modelling.

**Student meaning:** operate the interface, choose tools correctly, create precise geometry and edit drawings efficiently.

### Course Objective 2

To learn the basics of 2D modelling of engineering components.

**Student meaning:** convert dimensions and geometry into a complete, readable engineering drawing.

## Prerequisites

| Topic                          | Course name          | Semester/level   |
|--------------------------------|----------------------|------------------|
| Basics of engineering graphics | Engineering Graphics | Semester 1       |
| Geometry                       | Mathematics          | Secondary School |

## COURSE OUTCOMES

# What the student must demonstrate

| CO  | Official outcome                                           | Hours | Cognitive level | Evidence in this handbook                                               |
|-----|------------------------------------------------------------|-------|-----------------|-------------------------------------------------------------------------|
| CO1 | Identify the tools in CAD software.                        | 9     | Applying        | Interface, setup, coordinates, precision, draw/modify/annotation tools. |
| CO2 | Develop and draw figures using CAD software.               | 12    | Applying        | Six dimensioned 2D construction exercises.                              |
| CO3 | Sketch and practice isometric drawings using CAD software. | 12    | Applying        | Four isometric exercises and isoplane workflow.                         |
| CO4 | Construct two dimensional figures using CAD software.      | 6     | Applying        | Two complete engineering drawing sheets.                                |
| —   | Lab Exam                                                   | 6     | —               | Two three-hour model lab examinations.                                  |

## CO-PO mapping

| CO  | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | Interpretation                  |
|-----|-----|-----|-----|-----|-----|-----|-----|---------------------------------|
| CO1 | —   | —   | 3   | —   | —   | —   | —   | Strongly mapped to PO3.         |
| CO2 | 3   | —   | —   | 3   | —   | —   | —   | Strongly mapped to PO1 and PO4. |
| CO3 | 3   | —   | —   | 3   | —   | —   | —   | Strongly mapped to PO1 and PO4. |
| CO4 | 3   | —   | —   | —   | —   | —   | —   | Strongly mapped to PO1.         |

### Mapping scale

3 - Strongly mapped, 2 - Moderately mapped, 1 - Weakly mapped.

# Official hours and minimum drawing count

| Module outcome | Description                                         | Hours | Level         | Minimum evidence                          |
|----------------|-----------------------------------------------------|-------|---------------|-------------------------------------------|
| M1.01          | Study basic tools of CAD software and drawings.     | 3     | Understanding | Interface and tool identification record. |
| M1.02          | Apply software tools and draw the sketches.         | 6     | Applying      | Precision setup and guided sketches.      |
| M2.01          | Develop and draw figures using CAD.                 | 12    | Applying      | At least 6 figures.                       |
| Lab Exam 1     | Practical examination after Module 2.               | 3     | —             | One complete precision drawing task.      |
| M3.01          | Develop and draw isometric figures using CAD.       | 12    | Applying      | At least 4 figures.                       |
| M4.01          | Develop and draw two dimensional figures using CAD. | 6     | Applying      | At least 2 figures.                       |
| Lab Exam 2     | Practical examination after Module 4.               | 3     | —             | Isometric/2D complete drawing task.       |

## Hour check

$9 + 12 + 3 + 12 + 6 + 3 = 45$  periods, exactly matching the semester allocation.

# CAD Tools, Workspace and Precision Drafting

M1.01: Study basic tools and drawings. M1.02: Apply software tools and draw sketches.

## 1. CAD workspace

### Drawing area

The model space where geometry is created at full scale.

### Command line

Shows prompts, options and entered values. Read it continuously.

### Ribbon/toolbars

Groups Draw, Modify, Annotation, Layers and Properties tools.

### Status bar

Controls GRID, SNAP, ORTHO, POLAR, OSNAP, OTRACK and isometric drafting.

### Model and layout

Model space holds full-size geometry; layout space prepares paper and plot scale.

### Navigation

ZOOM and PAN change the view without changing object dimensions.

**Malayalam Support:** Model space-□ object actual size-□ draw □□□□□□. Zoom/Pan view □□□□□□ □□□□□□; drawing size □□□□□□□□.

## 2. Start every drawing correctly

1 Create a new drawing using a suitable metric template.

2 Run **UNITS**; choose Decimal and millimetres where required.

3 Save immediately with a meaningful name such as **2038\_M2\_Figure01.dwg**.

4 Create layers for OBJECT, CENTRE, HIDDEN, DIMENSION, CONSTRUCTION and TEXT.

5 Set OSNAP modes: Endpoint, Midpoint, Centre, Intersection, Quadrant and Tangent as needed.

6 Use ZOOM Extents after major stages and save repeatedly.

### 3. Coordinate entry

| Method             | Syntax                     | Use                                       | Example              |
|--------------------|----------------------------|-------------------------------------------|----------------------|
| Absolute Cartesian | x,y                        | Point measured from origin.               | 50,30                |
| Relative Cartesian | @Δx,Δy                     | Point measured from previous point.       | @80,0                |
| Relative polar     | @distance<angle            | Length and direction from previous point. | @60<30               |
| Direct distance    | Point cursor + type length | Fast orthogonal/polar drawing.            | Move right, type 100 |

**Malayalam Support:** @ □□□□□□ coordinate previous point-□□ reference □□□□□□. @50,0 □□□□□□ □□□□□□ point-□ □□□□□□ 50 mm right □□□□□□□□.

### 4. Precision controls

#### ORTHO

Restricts cursor movement to horizontal and vertical directions.

#### POLAR

Tracks selected angles such as 30°, 45° or 90°.

#### OSNAP

Locks the cursor to exact geometric points; never estimate visually.

#### Object tracking

Projects temporary alignment paths from snapped points.

#### Common mistake

Clicking near an endpoint is not the same as snapping to it. Confirm the OSNAP marker and tooltip before clicking.



## 5. Core draw and modify tools

| Group      | Commands                                                                                 | Purpose                                                  |
|------------|------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Draw       | LINE, PLINE, CIRCLE, ARC, RECTANGLE, POLYGON, ELLIPSE                                    | Create primary geometry.                                 |
| Modify     | MOVE, COPY, ROTATE, MIRROR, OFFSET, TRIM, EXTEND, FILLET, CHAMFER, SCALE, STRETCH, ARRAY | Edit and repeat geometry.                                |
| Properties | LAYER, MATCHPROP, PROPERTIES                                                             | Control colour, line type, line weight and organisation. |
| Annotation | TEXT/MTEXT, DIM, MLEADER, HATCH                                                          | Add dimensions, notes, leaders and section patterns.     |
| View/file  | ZOOM, PAN, SAVE, SAVEAS, PLOT                                                            | Navigate, protect and output the work.                   |

## 6. Drawing standards refresher

| Layer        | Suggested line type   | Typical use                    |
|--------------|-----------------------|--------------------------------|
| OBJECT       | Continuous, thick     | Visible outlines.              |
| CENTRE       | Centre line, thin     | Axes and centres of circles.   |
| HIDDEN       | Dashed, thin          | Invisible edges.               |
| DIMENSION    | Continuous, thin      | Dimension and extension lines. |
| CONSTRUCTION | Continuous, very thin | Temporary layout geometry.     |
| TEXT         | Continuous            | Notes and title block.         |

## 7. Guided starter sketch: centred rectangle with holes

GUIDED SKETCH

Target time: 20 minutes

100 × 60 plate with four Ø10 corner holes

Plate  
100 × 60

Hole centres  
15 from adjacent edges

Hole size  
Ø10

Tools  
RECTANGLE, OFFSET,  
CIRCLE, COPY

Suggested sequence: RECTANGLE → OFFSET 15 from four sides → CIRCLE R5 at first intersection → COPY to other intersections → erase/trim construction lines → DIM.

## Module 1 practice questions

### VERY SHORT

Differentiate model space and layout space.

### COMMAND

Write the relative polar coordinate entry for a 75 mm line at 30°.

### APPLICATION

List the OSNAP modes needed to draw a line tangent to two circles.

### PROCEDURE

Write the correct startup procedure for a metric engineering drawing.

### ERROR ANALYSIS

A drawing looks aligned but gaps appear when zoomed. Identify likely causes and corrections.

### PRACTICAL

Create the 100 × 60 plate, apply dimensions and save it with the specified file name.

# Six Foundational CAD Figures

The official syllabus requires at least six figures. The following exercises progress from simple offsets to tangent geometry, arrays and complete dimensioning.

---

**FIGURE 1**

Commands: RECTANGLE, FILLET, CIRCLE, OFFSET, TRIM

## Slotted mounting plate

**Overall**

120 × 70

**Corner radius**

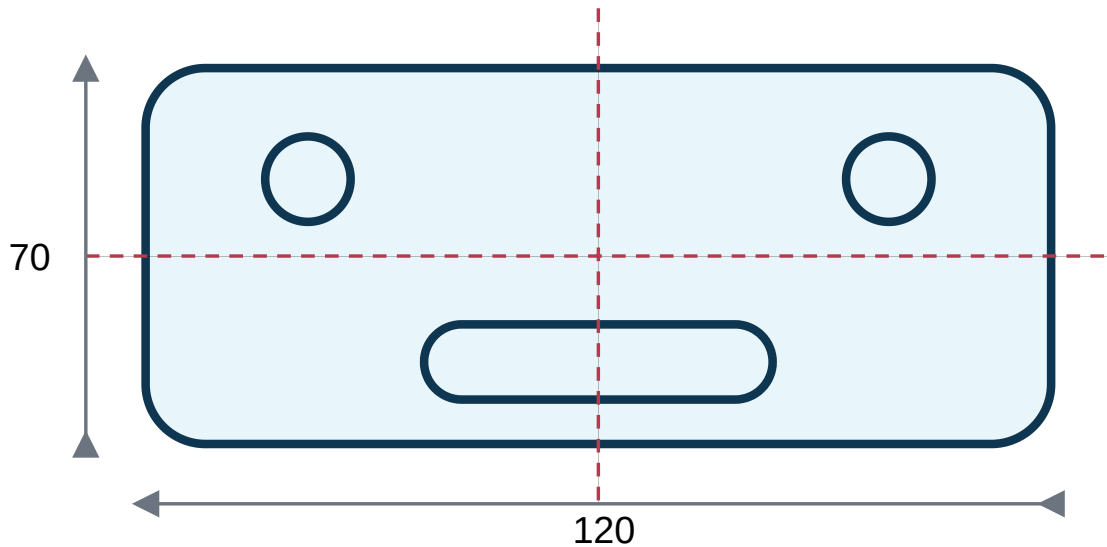
R10

**Holes**

2 × Ø12

**Slot**

50 × 14



Suggested construction: outer rectangle → fillet corners → centre lines → two holes → slot from two circles and tangent lines → trim → dimension.

- 1 Draw a 120 × 70 rectangle.
- 2 FILLET all corners with radius 10.
- 3 Create horizontal and vertical centre lines.
- 4 Locate hole centres and draw Ø12 circles.
- 5 Create the slot using two R7 circles 50 mm centre-to-centre and tangent lines; TRIM inner arcs.
- 6 Apply dimensions and centre marks.

**FIGURE 2**

Commands: CIRCLE, ARRAYPOLAR, CENTERMARK, DIMDIAMETER

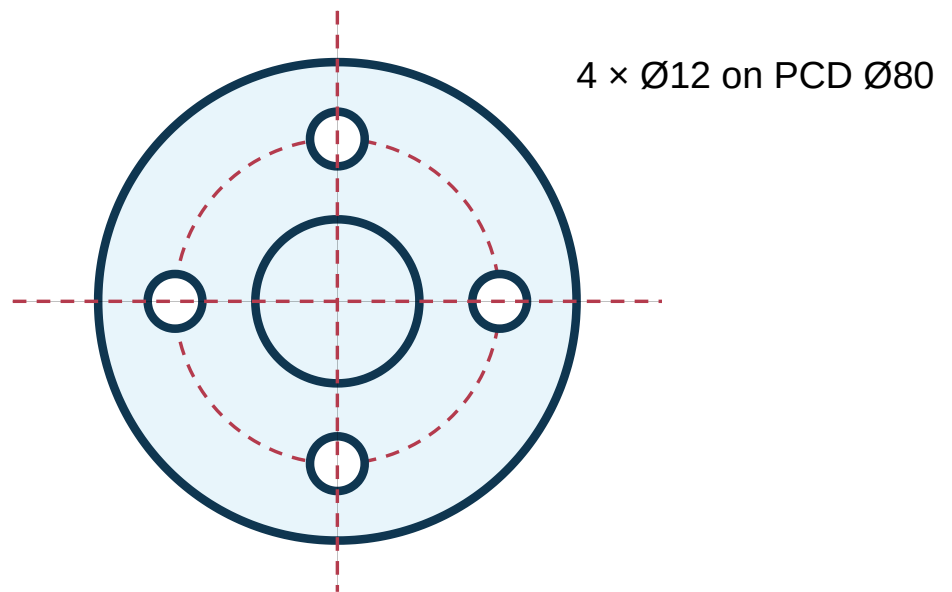
## Four-hole circular flange

**Outer**  
Ø120

**Bore**  
Ø40

**Pitch circle**  
Ø80

**Bolt holes**  
4 × Ø12



Draw one bolt hole on the pitch circle, then create a four-item polar array around the flange centre.

- 1 Draw concentric Ø120, Ø80 construction and Ø40 circles.
- 2 Place one Ø12 hole at the top quadrant of the pitch circle.
- 3 Use ARRAYPOLAR: centre = flange centre, items = 4, fill angle = 360°.
- 4 Place centre marks and diameter dimensions.

**FIGURE 3**

Commands: CIRCLE, LINE TAN, FILLET, TRIM

## Tangent cam profile

**Base circle**

Ø60

**Top circle**

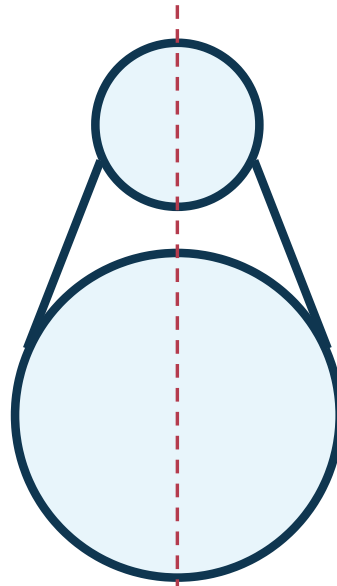
Ø30

**Centre distance**

70 vertical

**Sides**

Common tangents



Use TAN snap

The profile is built from two circles joined by common external tangents; unwanted arcs are trimmed.

- 1 Draw the two circles on a common vertical centre line.
- 2 Use LINE with Tangent OSNAP from the small circle to the large circle on each side.
- 3 TRIM internal circle portions and retain a closed cam outline.
- 4 Add centre marks and dimensions.

**FIGURE 4**

Commands: PLINE, OFFSET, CHAMFER, FILLET, MIRROR

## Stepped support profile

**Base**

140 × 20

**Web**

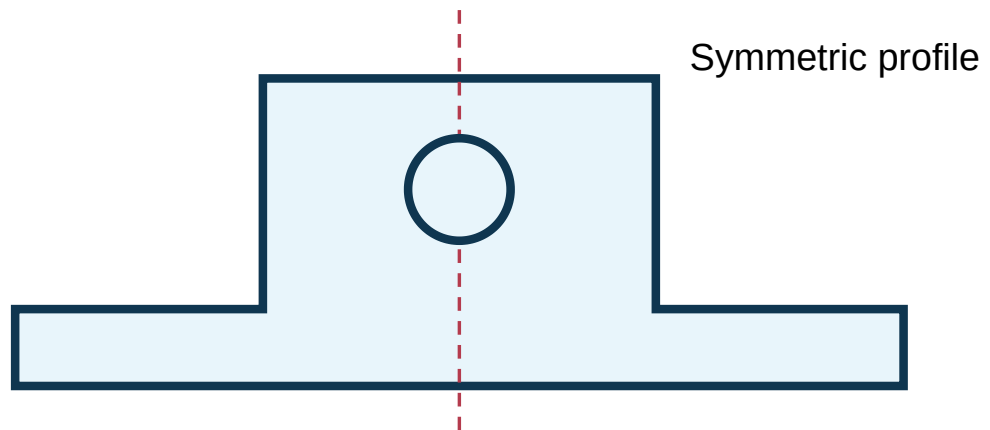
60 high

**Top**

80 × 20

**Corner**

R8 / 10×10 chamfer



Draw half of the symmetric outline accurately and MIRROR about the vertical centre line.

- 1 Draw the base and vertical centre line.
- 2 Use PLINE for one half of the stepped outline.
- 3 Apply required fillet/chamfer.
- 4 MIRROR the completed half, then draw the central hole.

**FIGURE 5**

Commands: RECTANGLE, CIRCLE, LINE, TRIM, HATCH

## Keyhole plate

**Plate**

100 × 80

**Upper circle**

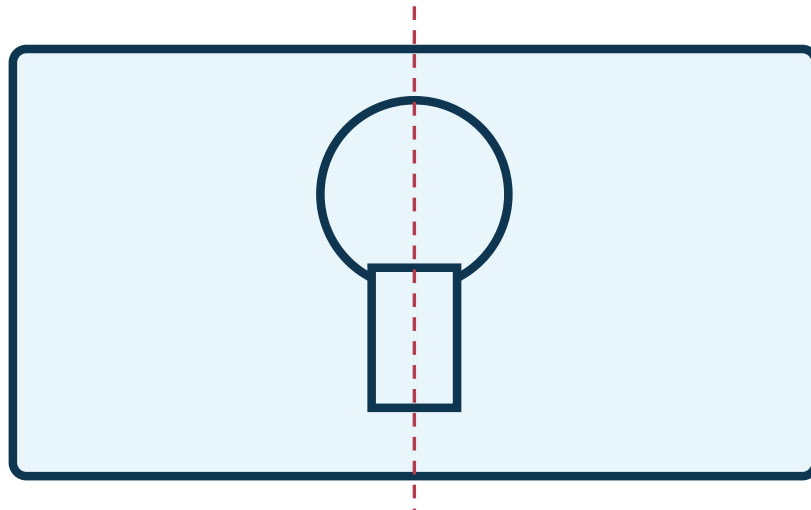
Ø30

**Stem**

14 wide × 30

**Position**

Centred



Create the circular opening and rectangular stem, then TRIM their overlapping boundary to form one keyhole opening.

- 1 Draw the outer plate and centre lines.
- 2 Draw Ø30 at the specified centre.
- 3 Draw a 14 mm wide centred stem below the circle.
- 4 TRIM the common circle segment and internal top line of the stem.



**FIGURE 6**

Commands: RECTANGLE, OFFSET, COPY, ARRAYRECT, TEXT, DIM

## Electrical panel mounting layout

**Panel**

180 × 120

**Border**

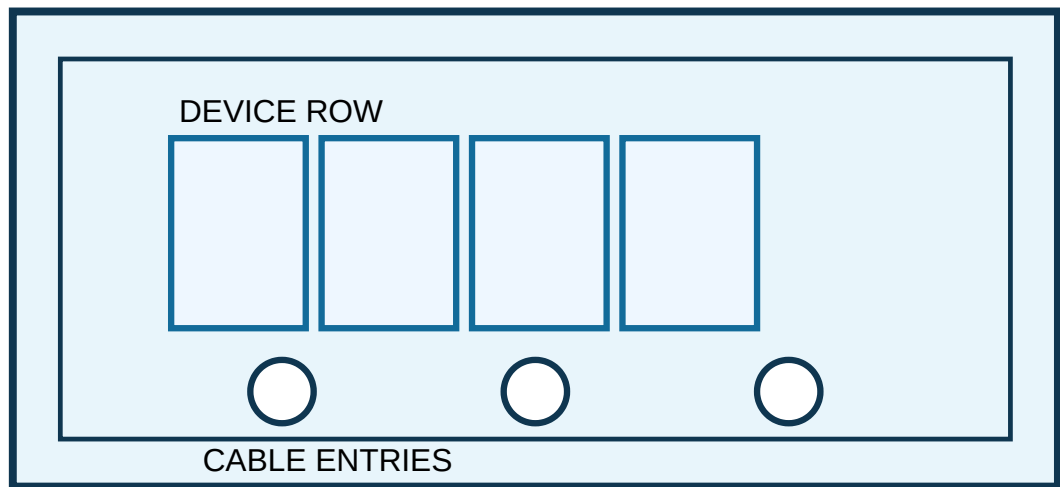
10 internal

**DIN devices**

4 modules 25 × 50

**Cable entry**

3 × Ø20



A practical layout exercise combining borders, repeated components, holes, labels and dimensions.

- 1 Draw panel outline and OFFSET 10 for the inner border.
- 2 Draw one device rectangle, then COPY or ARRAYRECT for four modules.
- 3 Draw one Ø20 cable entry and array/copy to three positions.
- 4 Add labels and overall/spacing dimensions on separate layers.

### Module 2 completion evidence

Submit six separate DWG files or one organised workbook drawing containing all six figures, with correct layers, dimensions and file names.

## Module 2 practice questions

### PROCEDURE

Write a command sequence for the four-hole flange.

### ERROR ANALYSIS

A polar array rotates the hole incorrectly. Explain which array settings must be checked.

### GEOMETRY

Explain why Tangent OSNAP is essential for Figure 3.

### PRACTICAL

Create any one of the six figures from dimensions without tracing the sample.

### EDITING

Compare OFFSET, COPY and ARRAY for repeated geometry.

### PLOT

Prepare Figure 6 for A4 plotting with readable dimensions.

# Isometric Drawing Using CAD

The official syllabus requires at least four isometric figures. These exercises build from a box to an engineering machine outline.

## 1. Isometric drafting essentials

### Isometric axes

One vertical axis and two receding axes at 30° to the horizontal.

### Isoplane

Left, Top and Right isoplanes control cursor orientation and isocircle plane.

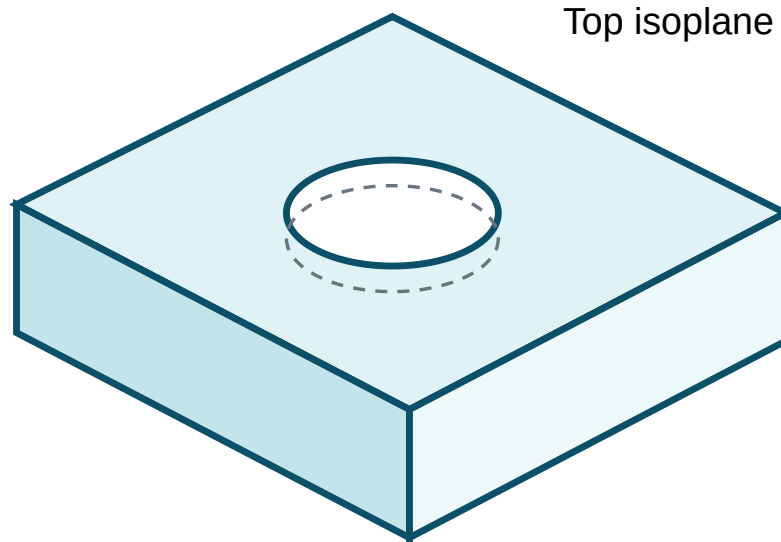
### Isocircle

Circles appear as ellipses; use ELLIPSE → Isocircle in isometric snap mode.

**Typical setup:** SNAP → Style → Isometric | toggle isoplane with **F5** or **Ctrl+E** | use ORTHO and POLAR.

**Malayalam Support:** Isometric mode- F5 Left/Top/Right isoplane . Circle draw normal CIRCLE, ELLIPSE → Isocircle .

## Rectangular block with through hole

**Block** $80 \times 50 \times 40$ **Hole** $\varnothing 20$  through top**Tools**LINE, OFFSET, ELLIPSE  
Isocircle, TRIM

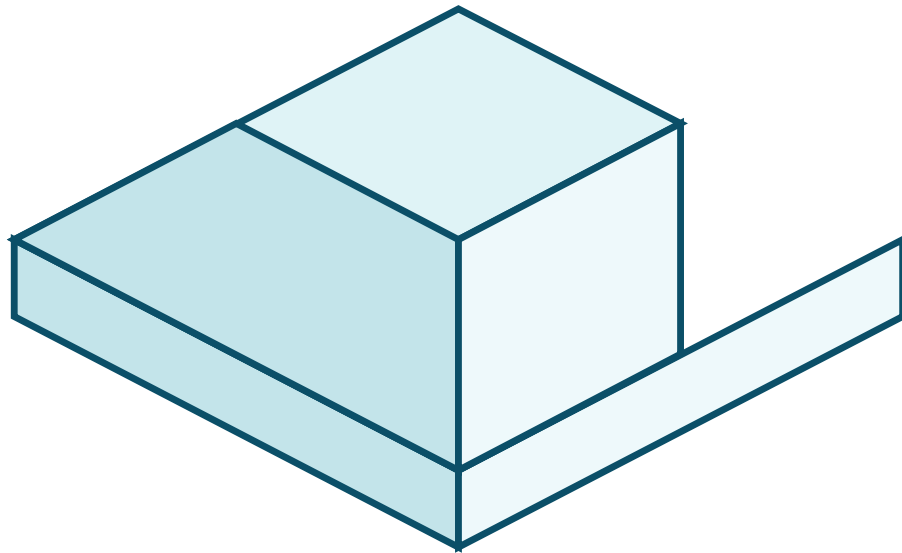
Construct the isometric box first. Locate the hole centre on the top face and create an isocircle.

- 1 Draw the three isometric axes from one corner.
- 2 Create the  $80 \times 50 \times 40$  bounding box.
- 3 Locate the top-face centre with diagonals/construction offsets.
- 4 Select Top isoplane and create  $\varnothing 20$  isocircle.
- 5 Add hidden lower ellipse only if the drawing convention/instructor requires it.

## Stepped block

**Overall** $90 \times 60 \times 60$ **Step**45 length  $\times$  30 height**Tools**

LINE, OFFSET, TRIM



Use an overall box, then remove the upper front portion conceptually. Construction-box thinking reduces errors.

- 1 Draw the complete  $90 \times 60 \times 60$  bounding box lightly.
- 2 Mark the step length and height on the correct axes.
- 3 Draw the new visible edges and TRIM construction edges.
- 4 Check that all receding lines remain parallel to the isometric axes.

## U-shaped bracket

**Outer** $90 \times 50 \times 60$ **Wall**

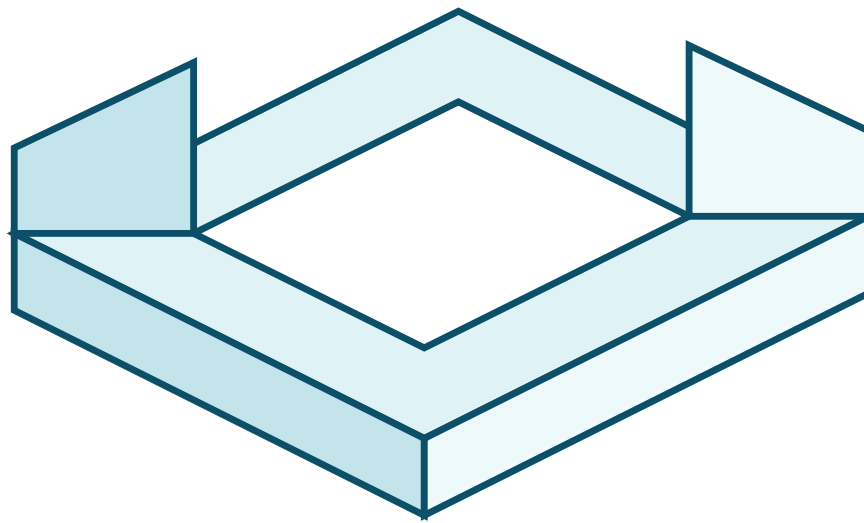
12

**Base**

12

**Tools**

LINE, OFFSET, COPY, TRIM



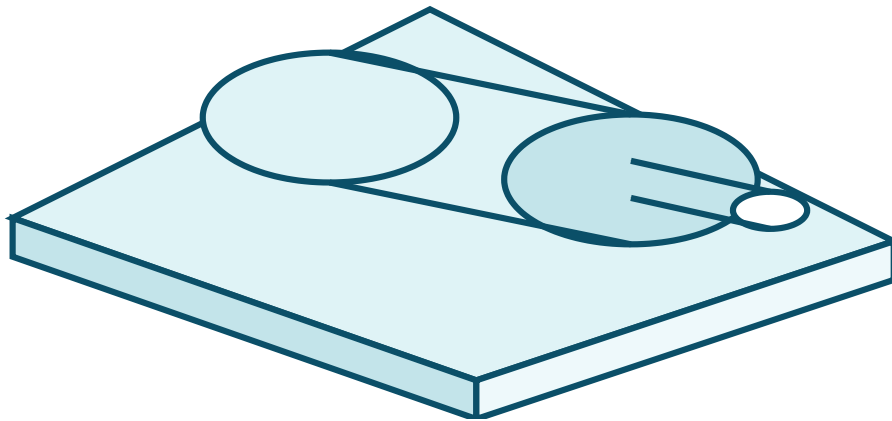
Build the base slab, then two equal side walls. Use offsets to maintain constant wall thickness.

- 1 Draw the base plate as an isometric slab.
- 2 OFFSET/construct 12 mm wall thickness on left and right.
- 3 Raise both walls to 60 mm height.
- 4 Remove hidden/construction geometry and verify symmetry.

## Simplified electric motor

**Body** $\text{Ø}60 \times 100$ **Shaft** $\text{Ø}16 \times 35$ **Base** $120 \times 55 \times 10$ **Tools**ELLIPSE Isocircle, LINE,  
OFFSET, TRIM

Simplified training form



A simplified training model, not a manufacturing drawing. It integrates isocircles, cylindrical form, shaft and base.

- 1 Draw the base plate.
- 2 Use the correct side isoplane to draw the motor-body end isocircles.
- 3 Join corresponding tangent points to form the cylinder.
- 4 Repeat for the shaft and add simple mounting feet.
- 5 Clean up hidden overlaps and add a descriptive title.

### Isometric common mistakes

Using normal circles; changing to the wrong isoplane; measuring along the screen instead of isometric axes; allowing receding lines to drift away from  $30^\circ$ ; trimming visible edges needed for the final form.

## Module 3 practice questions

### VERY SHORT

Name the three isoplanes.

### COMMAND

Explain how to draw a circle on an isometric top face.

### APPLICATION

Write a procedure for creating a cylindrical object in isometric CAD.

### PRACTICAL

Draw any one of the four isometric exercises from dimensions.

### ERROR ANALYSIS

An isocircle appears on the wrong plane. State the correction.

### VISUALISATION

Describe how a bounding box helps create a stepped block.



# Two Complete Two-Dimensional Engineering Drawings

The official syllabus requires at least two figures. Unlike the shorter Module 2 exercises, these are complete sheets with layers, annotation, title, dimensions and plot preparation.

## COMPLETE 2D FIGURE 1

### Drawing sheet workflow

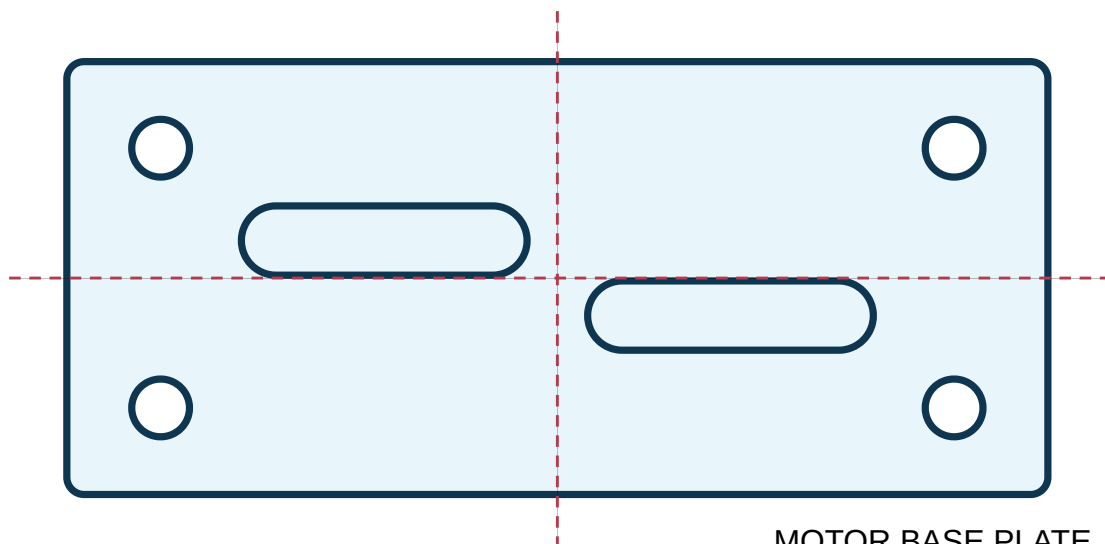
## Motor base plate

**Overall**  
200 × 120

**Corner holes**  
4 × Ø14, 20 from edges

**Motor slots**  
2 × 70 × 16

**Centre distance**  
80 between slots



A complete plate drawing with visible outlines, centre lines, holes, slots, dimensions, note and title.

- 1 Create layer set and assign line types.
- 2 Draw 200 × 120 outer profile and centre lines.
- 3 OFFSET 20 from edges to locate four hole centres; draw Ø14 holes.
- 4 Construct two 70 × 16 slots at the specified centre spacing.
- 5 Add centre marks, overall dimensions, hole note and slot dimensions.
- 6 Add title and inspect plot preview on A4 landscape.

## Electrical terminal mounting plate

**Plate**

180 × 100

**Terminal row**

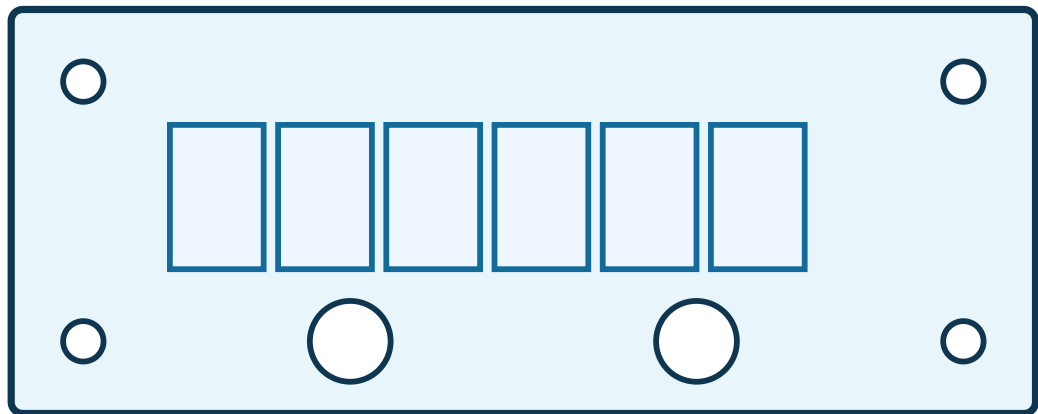
6 × 20 × 30

**Fixing holes**

4 × Ø8

**Cable entries**

2 × Ø24



TERMINAL MOUNTING PLATE

The completed sheet should show arrayed terminal modules, fixing holes, cable entries, dimensions and component labels.

- 1 Draw outer plate and inner construction margins.
- 2 Create one terminal block rectangle and use ARRAYRECT/COPY for six.
- 3 Locate four Ø8 fixing holes with offsets.
- 4 Create two Ø24 cable-entry holes on the lower centre line.
- 5 Add component numbers T1-T6, diameter notes and overall/spacing dimensions.
- 6 Prepare title block and plot preview.

## Dimensioning and plotting checklist

### Geometry

- ✓ Closed profiles

- ✓ No duplicate lines
- ✓ Exact centres and tangencies

### Annotation

- ✓ No repeated dimensions
- ✓ Diameter/radius symbols correct
- ✓ Text height readable

### Plot

- ✓ Correct paper size/orientation
- ✓ Line weights visible
- ✓ Drawing centred

## Module 4 practice questions

#### PROCEDURE

Prepare a layer and dimension plan for the motor base plate.

#### APPLICATION

Explain how ARRAYRECT reduces work in the terminal plate.

#### QUALITY

List checks to perform before plotting a complete 2D drawing.

#### PRACTICAL

Complete either figure with title, dimensions and plot setup.

# Purpose, syntax and practical use

Command names follow common AutoCAD terminology. Other CAD packages may use different icons or names, but the geometric operations remain similar.

## Keyboard and workflow shortcuts

| Key/action    | Purpose                               | Lab use                                  |
|---------------|---------------------------------------|------------------------------------------|
| Esc           | Cancel current command.               | Recover from an incorrect option safely. |
| Enter / Space | Accept option or repeat last command. | Faster repeated operations.              |
| F3            | Toggle OSNAP.                         | Precision point selection.               |
| F8            | Toggle ORTHO.                         | Horizontal/vertical drawing.             |
| F10           | Toggle POLAR.                         | Angle tracking.                          |
| F5 / Ctrl+E   | Cycle isoplanes.                      | Isometric drawing.                       |
| Ctrl+S        | Save.                                 | Protect work frequently.                 |
| Ctrl+Z        | Undo.                                 | Reverse the last action.                 |

### File safety

Save after setup, after main geometry, after annotation and before plotting. Keep a backup copy. Do not overwrite another student's file.

# Record format, submission checklist and self-assessment

## Recommended record format for every exercise

| Record item                 | What to write                                           |
|-----------------------------|---------------------------------------------------------|
| Experiment number and title | Example: M2-F3 - Tangent cam profile.                   |
| Aim                         | One sentence stating the required CAD drawing.          |
| Software/tools              | CAD package and important commands.                     |
| Given data                  | Dimensions, quantities and assumptions.                 |
| Procedure                   | Ordered command and construction steps.                 |
| Result                      | Drawing completed, checked, saved and plotted/exported. |
| File evidence               | DWG/DXF filename and screenshot/print.                  |
| Learning reflection         | One error faced and how it was corrected.               |

## Minimum semester drawing register

| No. | Exercise                      | Official minimum category | File name           | Checked                  |
|-----|-------------------------------|---------------------------|---------------------|--------------------------|
| 1   | Starter plate with four holes | M1 sketch                 | 2038_M1_Starter.dwg | <input type="checkbox"/> |
| 2   | Slotted mounting plate        | M2 Figure 1               | 2038_M2_F1.dwg      | <input type="checkbox"/> |
| 3   | Circular flange               | M2 Figure 2               | 2038_M2_F2.dwg      | <input type="checkbox"/> |
| 4   | Tangent cam                   | M2 Figure 3               | 2038_M2_F3.dwg      | <input type="checkbox"/> |
| 5   | Stepped support               | M2 Figure 4               | 2038_M2_F4.dwg      | <input type="checkbox"/> |
| 6   | Keyhole plate                 | M2 Figure 5               | 2038_M2_F5.dwg      | <input type="checkbox"/> |
| 7   | Panel mounting layout         | M2 Figure 6               | 2038_M2_F6.dwg      | <input type="checkbox"/> |
| 8   | Block with hole               | M3 Isometric 1            | 2038_M3_I1.dwg      | <input type="checkbox"/> |
| 9   | Stepped block                 | M3 Isometric 2            | 2038_M3_I2.dwg      | <input type="checkbox"/> |
| 10  | U bracket                     | M3 Isometric 3            | 2038_M3_I3.dwg      | <input type="checkbox"/> |
| 11  | Electric motor form           | M3 Isometric 4            | 2038_M3_I4.dwg      | <input type="checkbox"/> |
| 12  | Motor base plate              | M4 2D Figure 1            | 2038_M4_D1.dwg      | <input type="checkbox"/> |
| 13  | Terminal mounting plate       | M4 2D Figure 2            | 2038_M4_D2.dwg      | <input type="checkbox"/> |

# Drawing quality checklist

## Accuracy

- ✓ All specified dimensions used
- ✓ No visual guessing
- ✓ Centres and tangencies exact
- ✓ Closed profiles where required

## CAD method

- ✓ Correct commands selected
- ✓ Layers used logically
- ✓ No unnecessary duplicate entities
- ✓ File saved with correct name

## Presentation

- ✓ Dimensions readable
- ✓ Line types visible
- ✓ Title and labels present
- ✓ Plot preview checked

## Viva practice

### VIVA

Why is OSNAP more reliable than visual selection?

### VIVA

Differentiate OFFSET and COPY.

### VIVA

Why should dimensions be placed on a separate layer?

### VIVA

What is the purpose of plot preview?

### VIVA

How is an isocircle created?

### VIVA

What happens if the wrong units are used?

# Two model three-hour lab examinations

The syllabus allocates Lab Exam 1 after Module 2 and Lab Exam 2 after Module 4, three hours each. The following are fresh practice models, not an official mark scheme.

## MODEL LAB EXAM 1

Time: 3 hours

### Precision 2D CAD figure

**Task:** Draw a  $160 \times 100$  mounting plate with R12 rounded corners, four  $\varnothing 12$  holes located 20 mm from adjacent edges, a central  $\varnothing 40$  opening, and two horizontal slots  $50 \times 12$  placed symmetrically above and below the centre. Use separate object, centre and dimension layers. Add all essential dimensions and prepare A4 landscape plot preview.

#### Expected submission

- ✓ DWG/DXF source file
- ✓ Correct layer structure
- ✓ Fully dimensioned drawing
- ✓ Plot preview or PDF

#### Planning

Units, file and layers set before drawing.

#### Geometry

Dimensions, centres, radii and slots accurate.

#### Technique

Appropriate use of OFFSET, COPY/MIRROR, FILLET and TRIM.

#### Presentation

Readable annotation and plot setup.

## Isometric equipment support

**Task:** Create an isometric drawing of a U-shaped equipment support having an overall base  $100 \times 60 \times 12$ , two side walls 12 thick and 70 high, and one  $\varnothing 20$  through hole in each side wall at equal height. Add a suitable title and prepare plot preview.

### Expected submission

- ✓ Correct isometric snap and isoplane use
- ✓ Base and equal wall thickness
- ✓ Two correctly oriented isocircles
- ✓ Clean visible outline and title

#### Visualisation

Correct interpretation of dimensions.

#### Isometric method

Axes, isoplanes and isocircles used correctly.

#### Accuracy

Constant thickness and aligned holes.

#### Output

Saved file and clear plot preview.

## Lab exam strategy

- 1 Spend the first 10 minutes reading, listing dimensions and planning commands.
- 2 Set units, save and create layers before geometry.
- 3 Create construction geometry first; do not dimension incomplete geometry.
- 4 Use symmetry, arrays and offsets to reduce repeated manual work.
- 5 Reserve the final 25 minutes for checking, annotation, title, save and plot preview.



# Open-ended group experiment

The official syllabus specifies groups of 3-5, no duplication between groups, a separate report and 15 marks. The given example is an isometric drawing of an electric machine using CAD software.

## Official conditions

Work in a group of 3-5. Select a unique experiment not duplicated by another group. Prepare and submit a separate report. The activity is intended for continuous internal evaluation with a score of 15 marks.

## Suggested project themes

### Electric machine

Develop a simplified isometric drawing of a motor or generator with base, body, shaft and terminal box.

### Renewable-energy frame

Model a simplified solar-panel mounting frame or battery rack.

### Electrical panel

Create a 2D panel layout with device rows, cable entries and mounting dimensions.

### EV component support

Develop a mounting bracket or battery-module tray using 2D and isometric views.

### Workshop fixture

Draw a simple jig, clamp or equipment support.

### Campus utility

Prepare a dimensioned equipment-base or cable-tray support drawing.

## Project workflow

**1** Choose a unique object and obtain instructor approval.

**2** Measure or define safe simplified dimensions.

- 3 Prepare a freehand concept sketch and drawing plan.
- 4 Assign group roles: geometry, layers/annotation, verification, report and presentation.
- 5 Create CAD drawing, review dimensions and produce plot/PDF.
- 6 Write the report and present the method, difficulties and corrections.

## Separate report structure

| Section            | Required content                                                 |
|--------------------|------------------------------------------------------------------|
| Title page         | Course, project title, group members, register numbers and date. |
| Objective          | What the group intended to draft/model.                          |
| Object description | Function, main parts and simplified assumptions.                 |
| Dimensions/data    | Dimension table and source of data.                              |
| CAD plan           | Layers, commands and drawing sequence.                           |
| Procedure          | Step-by-step execution with screenshots.                         |
| Final output       | Dimensioned 2D/isometric drawing and plot.                       |
| Challenges         | Errors, corrections and team learning.                           |
| Conclusion         | Result and possible improvement.                                 |

### Suggested 15-mark rubric (instructor adaptable)

Planning and uniqueness: 3; CAD accuracy: 5; use of tools/layers: 3; report: 2; teamwork/presentation: 2. This is a suggested handbook rubric, not an official distribution stated in the syllabus.

# Complete answers and procedure-based solutions

This section contains answers to module questions, viva questions and full model lab-exam procedures.

## Module 1 answers

**Model space vs layout space:** Model space is used to create full-size geometry. Layout space represents the paper sheet and viewports used for plotting at a selected scale.

**75 mm at 30°:** Enter `@75<30` after selecting the start point.

**Tangent line between circles:** Enable Tangent OSNAP and select the first circle at the tangent marker, then the second circle at its tangent marker.

**Metric startup:** New drawing → UNITS Decimal/mm → SAVEAS → create layers → set line types → set OSNAP/ORTHO/POLAR → begin construction.

**Gaps after zooming:** Likely causes are visual clicking without OSNAP, wrong snap mode, incomplete TRIM/EXTEND or duplicate objects. Turn on Endpoint/Intersection snaps, use EXTEND/TRIM, and inspect with zoom.

## Module 2 answers

### Four-hole flange procedure

1. Set units and layers.
2. Draw concentric circles Ø120, Ø80 and Ø40.
3. Use the Ø80 circle as a construction pitch circle.
4. At one quadrant, draw a Ø12 bolt hole.
5. ARRAYPOLAR around the common centre with 4 items and 360° fill.
6. Add centre marks and diameter/PCD note.
7. Freeze/delete construction circle only if required by the instructor.

**Array rotates holes incorrectly:** Check the array centre, item count, fill angle, association option and whether “rotate items” affects a non-circular object. For circular holes, centre location and item count are the critical settings.

**Why Tangent OSNAP:** It computes the exact contact point where a line touches a circle without crossing it. Visual placement cannot guarantee tangency.

**OFFSET vs COPY vs ARRAY:** OFFSET creates a parallel/concentric object at a fixed distance; COPY duplicates selected objects using displacement; ARRAY produces multiple copies in a rectangular, polar or path pattern.

**A4 plot preparation:** Open layout/plot, select A4 and orientation, choose printer/PDF, select plot area/layout, centre the drawing, set scale or Fit only when appropriate, apply line-weight plot style, preview, then plot.

## Module 3 answers

**Three isoplanes:** Left, Top and Right.

**Circle on top face:** Activate isometric snap, switch to Top isoplane, run ELLIPSE, select Isocircle, choose centre and enter radius/diameter.

### Cylindrical object procedure

1. Select the isoplane perpendicular to the cylinder axis.
2. Draw the near-end isocircle.
3. Project the cylinder length along the correct isometric axis.
4. Copy/draw the far-end isocircle.
5. Join corresponding extreme tangent points.
6. Trim or suppress hidden portions according to the required convention.

**Wrong isocircle plane:** Undo/delete it, cycle to the correct isoplane with F5/Ctrl+E, then recreate the isocircle.

**Bounding box method:** First draw the maximum overall length, width and height. Mark cuts/steps on the box edges, join corresponding points, then remove construction edges. This preserves proportions and axis directions.

## Module 4 answers

**Layer plan:** OBJECT continuous/thick; CENTRE centre line/thin; HIDDEN dashed/thin when needed; DIMENSION continuous/thin; CONSTRUCTION light; TEXT continuous. Draw geometry first, then dimensions.

**ARRAYRECT in terminal plate:** It creates equally spaced copies in rows/columns from one accurate terminal rectangle, reducing repeated measurement and maintaining uniform spacing.

**Pre-plot checks:** Units, overall size, duplicate/gapped entities, layer assignment, dimension completeness, text height, line-type scale, line weights, paper size, orientation, plot area, scale and preview.

## Viva answers

| Question                     | Model answer                                                                                                   |
|------------------------------|----------------------------------------------------------------------------------------------------------------|
| Why OSNAP?                   | It selects exact endpoints, centres, intersections, quadrants and tangencies, ensuring geometric accuracy.     |
| OFFSET vs COPY?              | OFFSET creates parallel/concentric geometry at a distance; COPY produces a duplicate at a chosen displacement. |
| Separate dimension layer?    | Dimensions can be controlled, hidden, printed and edited independently from object geometry.                   |
| Purpose of plot preview?     | It reveals clipping, scale, orientation, line-weight and readability problems before final output.             |
| How is an isocircle created? | Isometric snap → correct isoplane → ELLIPSE → Isocircle.                                                       |
| Wrong units effect?          | All entered distances and scale relationships become incorrect, causing an unusable engineering drawing.       |

## Complete Model Lab Exam 1 solution

### FULL PROCEDURE

### Expected method

#### 160 × 100 precision mounting plate

1. Start metric drawing; save as 2038\_LabExam1.dwg .
2. Create OBJECT, CENTRE, DIMENSION and CONSTRUCTION layers.
3. RECTANGLE from 0,0 to @160,100. FILLET radius 12 on all corners.
4. OFFSET 20 inward from left/right/top/bottom construction references. Their four intersections are hole centres.
5. Draw one CIRCLE diameter 12 and COPY to the other three centres.
6. Draw centre lines through plate centre. Draw central CIRCLE diameter 40.
7. For the upper slot, mark two centres 50 mm apart on a horizontal line. Draw two R6 circles, join with tangent horizontal lines and TRIM inner arcs. Place the slot symmetrically above centre.
8. MIRROR or COPY the slot to an equal distance below centre.
9. Remove unnecessary construction lines; apply centre marks.
10. Dimension overall size, R12, 4×Ø12, Ø40, slot 50×12 and location/spacing dimensions.
11. Add title, save, open A4 landscape plot preview and check line weights/readability.

**Expected final view:** one symmetric rounded rectangular plate with four corner holes, central circular opening and two equal horizontal slots.

# Complete Model Lab Exam 2 solution

## FULL PROCEDURE

## Expected method

### Isometric U-shaped equipment support

1. Start metric drawing, save as `2038_LabExam2.dwg`, enable isometric snap.
2. Draw the  $100 \times 60$  top outline of the base along isometric axes.
3. Project 12 mm downward and close the base slab.
4. On the left and right edges, construct 12 mm wall thickness.
5. From the top of the base, project both walls to 70 mm height and close their top faces.
6. Mark equal hole centres on both walls using offsets from base and wall edges.
7. Switch to the correct left/right isoplane and create  $\varnothing 20$  isocircles.
8. Trim construction lines and retain clean visible outlines.
9. Add a title; optional dimensions may be placed if required by the instructor.
10. Save and prepare plot preview.

**Expected final view:** a symmetric isometric U bracket with a 12 mm base, equal 12 mm side walls and aligned circular holes represented by correctly oriented isocircles.

# Textbooks and online resources from the syllabus

| Code | Book / author                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------|
| T1   | <i>Engineering Graphics with Auto CAD</i> - R. B. Choudary, Anuradha Publishers.                                   |
| T2   | K. C. John, <i>Engineering Graphics</i> , PHI Learning Private Limited.                                            |
| R1   | P. I. Varghese, <i>Engineering Graphics</i> , VIP Publishers.                                                      |
| R2   | Anilkumar K. N., <i>Engineering Graphics</i> , Adhyuth Narayan Publishers.                                         |
| R3   | N. D. Bhatt, <i>Engineering Drawing</i> , Charotar Publishing House, Anand, Gujarat, 2010; ISBN 978-93-80358-17-8. |
| R4   | M. B. Shah and B. C. Rana, <i>Engineering Drawing</i> , Pearson Publications.                                      |

# Online resources

| No. | Resource                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | How it supports the course                   |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| 1   | AutoCAD overview                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | General software background and terminology. |
| 2   | NPTEL course 112102101                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Engineering drawing/CAD learning support.    |
| 3   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Video demonstration.                         |
| 4   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Additional guided practice.                  |
| 6   | <div><b>Software note</b><br/>Interfaces and command options vary by software version. Use the installed laboratory versio checklist</div> <div><b>Before drawing</b><ul style="list-style-type: none"><li>✔ Question read completely</li><li>✔ Units set</li><li>✔ File saved</li><li>✔ Layers created</li><li>✔ OSNAP selected</li></ul></div> <div><b>Before submission</b><ul style="list-style-type: none"><li>✔ Geometry checked</li><li>✔ No duplicates/gaps</li><li>✔ Dimensions complete</li><li>✔ Title present</li><li>✔ Plot preview checked</li><li>✔ Final save completed</li></ul></div> |                                              |